

Alpha-Spending Functions

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Introduction

Alpha-spending functions (Lan & DeMets, 1983) allow interim analyses at unscheduled times while preserving overall Type I error. The spending function $f(t)$ specifies the cumulative alpha budget at information fraction $t \in [0, 1]$; nominal alpha at each analysis is the increment $f(t_k) - f(t_{k-1})$.

Prerequisites

Group-sequential designs; information fraction.

Theory

Common spending functions: - **OF-type**: $f(t) = 2 - 2\Phi(z_{\alpha/2}/\sqrt{t})$. Approximates the OF boundary. - **Pocock-type**: $f(t) = \alpha \log(1 + (e - 1)t)$. Approximates Pocock. - **Power family**: $f(t) = \alpha t^\rho$ with $\rho > 0$. - **Custom**: any non-decreasing f with $f(0) = 0, f(1) = \alpha$.

Flexibility: interim analyses can occur at arbitrary information fractions, re-solving the boundary each time.

Assumptions

Information fractions are known (approximately); stopping rule applied as specified; test statistic is normal.

R Implementation

```
library(rpact)

# OF spending, flexible information fractions
design <- getDesignGroupSequential(
  sided = 1, alpha = 0.025, beta = 0.2,
  typeOfDesign = "asOF",          # alpha-spending OF-type
  informationRates = c(0.4, 0.75, 1)
)

print(design$stageLevels)
plot(design, type = 1)

# Compare spending functions
f_of <- getDesignGroupSequential(sided = 1, alpha = 0.025,
                                typeOfDesign = "asOF",
                                informationRates = seq(0.1, 1, 0.1))
f_poc <- getDesignGroupSequential(sided = 1, alpha = 0.025,
                                  typeOfDesign = "asP",
                                  informationRates = seq(0.1, 1, 0.1))
cbind(OF = cumsum(f_of$stageLevels),
      Pocock = cumsum(f_poc$stageLevels))
```

Output & Results

Cumulative alpha at each information fraction for both spending families; OF delays alpha consumption, Pocock distributes it earlier.

Interpretation

“The alpha-spending design at information fractions 0.4, 0.75, 1.0 under OF-type spending allocated cumulative alpha of 0.001, 0.013, 0.025 respectively, enabling flexibility in interim timing without inflating Type I error.”

Practical Tips

- Alpha spending is the **standard** for modern confirmatory group-sequential trials.
- Re-estimate information fractions at each interim if accrual deviates from plan.
- Information is usually subject-count for continuous outcomes, event-count for time-to-event.
- Never spend more alpha than the planned cumulative function at the current information fraction.
- For futility, use separate **beta-spending** functions (non-binding boundaries are standard).

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale